

P12 Librarian

User Guide

Back up, browse, rename and restore the programs
of a Prophet 12 over MIDI SysEx

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Tested with Prophet 12 Keyboard, OS 1.4.2.3

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1 Introduction

P12 Librarian is a desktop app for managing the programs (patches) stored in a Dave Smith Instruments Prophet 12. It talks to the synth over MIDI SysEx and lets you:

- **back up** all User banks, or any selection, to standard `.syx` files;
- **browse and search** every program by name;
- **audition** patches from a file without storing them;
- **write** patches to any User slot, including into **Layer B** of an existing program;
- **rename** patches, and **clean up** names typed in capitals on the synth;
- **initialize** slots with the synth's Basic Patch.

Every write is protected the same way: the slots about to change are read and saved first, and every written program is read back from the synth and compared with what was sent (Section 9).

Use at your own risk. Writing to the synth replaces the programs stored there, and data can be lost. Keep your own backups. The app asks you to accept the disclaimer the first time it starts (Section 16).

1.1 Tested setup

Synthesizer	Prophet 12 Keyboard
Synth OS	1.4.2.3, the final release from DSI
Connection	USB, directly to the computer
Computer	Linux Mint 22.3 (Cinnamon, X11)

> **Note**
> The **Prophet 12 Module** is expected to work the same way, but has not been tested. Windows and macOS builds are possible with the same code but have not been tested either.

What was verified on the hardware above, beyond what the manual documents:

- where the program name is stored inside a program (the manual doesn't say);
- that writing, renaming and initializing store exactly what was sent (every write is verified);
- that lowercase letters in names display correctly;
- that the synth's own **Copy A -> B** copies a whole layer, which is what Layer B copying is based on.

2 Getting started

2.1 Settings on the synth

The Prophet 12 ignores SysEx until it's enabled. Press **Global** and set:

Global setting	Value	Why
10 MIDI SysEx Enable	On	Lets the synth receive and send SysEx.
11 MIDI SysEx Cable	USB	Use **MIDI Port** instead if you connect through a MIDI interface's DIN ports.

2.2 Connecting

Connect the Prophet 12 by USB and start the app. It looks for MIDI ports with "Prophet 12" in the name and connects automatically. The header then shows ● Connected.

If it doesn't connect, choose the ports in the **In** and **Out** lists, click **Connect**, and use ↻ to rescan after plugging the synth in. Connecting checks that a Prophet 12 really answers; if it doesn't, see Section 12.

2.3 The window

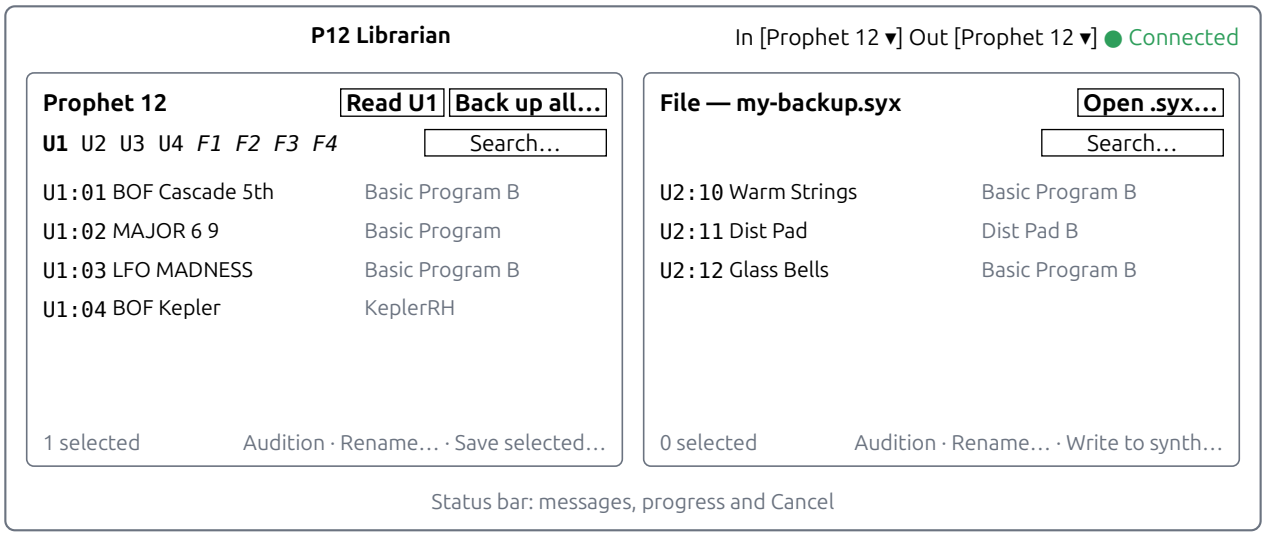


Figure 1: The main window: the synth on the left, an opened file on the right.

- **Header:** MIDI ports and connection status.
- **Prophet 12 pane (left):** the programs on the synth, one bank at a time. Tabs U1–U4 are the User banks; F1–F4 (in italics) are the read-only Factory banks. The small number on a tab is how many of its 99 slots have been read. Slots not read yet show "—".
- **File pane (right):** the patches in an opened `.syx` file.
- **Status bar:** what just happened, or the progress of a running operation.

Every button has a tooltip. If a button is greyed out, its tooltip says why.

2.4 Slots

A slot is written as bank and program number, as on the synth's panel: `U1:05` is User bank 1, program 5; `F3:99` is Factory bank 3, program 99. The User banks U1–U4 can be written; the Factory banks F1–F4 are read-only on the synth, and the app never writes to them.

Each program has two layers, A and B. The name in the **Name** column is Layer A's, the one the synth shows; the **Layer B** column shows the second layer's name (Section 6).

2.5 Selecting

Click	Select one patch.
`Ctrl` + click	Add or remove a patch.
`Shift` + click	Select a range.
Double-click	Audition the patch (Section 4.3).
Right-click	Menu with everything you can do with the selection.

3 Backing up

> **Important**
> Back up all User banks **before updating the synth's OS**. OS updates have been reported to change or erase user programs. Keep a copy of the backup somewhere other than the computer.

3.1 Reading programs

- **Read U1** (the name follows the open tab) reads the 99 programs of that bank, in about 3 seconds.
- **Read all User banks** reads all 396 User programs, in about 10 seconds.

- To read a Factory bank, open its tab and click **Read F1** (and so on).

A progress bar shows each program as it arrives. **Cancel** stops reading; programs read so far are kept.

3.2 Backing up everything

Back up all… (Ctrl+B, also in the **File** menu) asks where to save, reads all four User banks and saves them to one `.syx` file.

> **Example: a backup before an OS update**
>
> 1. Press Ctrl+B.
> 2. Keep the suggested name, e.g. `p12-backup-20261008-210500.syx`, or type your own, and click **Save**.
> 3. Wait about 10 seconds. The status bar says *"Saved all 396 User programs to …"*.
> 4. Copy the file to a USB stick or cloud storage.

3.3 Saving a selection

Select patches in either pane, then:

- **Save selected…** saves them into **one** `.syx` file;
- **Save as separate files…** saves **one file per patch** into a folder you choose, named after slot and name, e.g. `U1-05 BOF Cascade 5th.syx`.

The files are ordinary SysEx: other librarians and SysEx tools can read them, and each one remembers which slot it came from.

4 Writing patches to the synth

4.1 Opening a file

Open .syx… (Ctrl+O) loads a `.syx` file into the File pane. Files from this app, from the synth's own **Dump** functions and from other tools work. SysEx in the file that isn't a Prophet 12 program (for example the play lists in a **Dump All User Banks** file) is skipped and counted in the status bar.

4.2 Writing

There are two ways to write patches from the File pane:

- **Drag** them onto a slot in the Prophet 12 pane. Several patches fill consecutive slots from where you drop them; the slots that will be filled are outlined while you drag.
- Select them and click **Write to synth…**, then choose either the patches' **original locations** or **consecutive slots starting at** a slot you pick.

Either way, a confirmation lists every slot, the patch going there, and what it replaces. Nothing is written until you click **Write**.

> **Example: restoring one patch from a backup**
>
> 1. **Open .syx…** and choose your backup.
> 2. Type part of the name in the File pane's search box, e.g. `strings`.
> 3. Select the patch and click **Write to synth…**.
> 4. Leave **Original locations** selected and check that the **Replaces** column shows what you expect.
> 5. Click **Write**. The status bar confirms that the patch was written and verified.

Example: copying a patch to another slot on the synth
Patches in the Prophet 12 pane can be dragged too. Drag `U1:04` onto the row of `U4:50` to put a copy there, after the usual confirmation.

4.3 Audition

Double-click a patch, or select one and click **Audition**, to load it into the synth's **edit buffer** and play it from the keyboard. Nothing is stored: selecting a program on the synth loads its stored version again.

Important
Auditioning replaces the sound currently being edited on the synth. Save any unsaved edits on the synth first. To keep an auditioned sound, press **Write** on the synth and choose a slot.

5 Renaming

5.1 Rename

Select one or more patches and click **Rename…**, press F2, or right-click and choose **Rename…**.

Each patch gets an editable name field. Names can be up to 20 characters: letters (upper and lower case), digits, spaces and most punctuation. The synth can't show `* / : < > ? \ | ~`, so those are refused and the field turns red.

- In the **Prophet 12** pane, renaming stores the new name on the synth. Each slot is **read again first** and only its 20 name characters change, so edits you've made on the synth since reading it are kept. Factory patches can't be renamed; copy them to a User slot first.
- In the **File** pane, renaming changes the list in the app only. Save the patches or write them to the synth to keep the new names.

5.2 Cleanify names

Names typed on the synth are often in capitals. Select patches, right-click and choose **Cleanify names…**. The app suggests tidier names and shows them in the same dialog, where you can change any of them before confirming. Names that wouldn't change aren't listed.

Before	Suggested	Why
`BOF COOL PAD`	`Bof Cool Pad`	Each word in capitals gets one capital letter.
`DISTPAD`	`DistPad`	Run-together sound words are split.
`WARMSTRINGS`	`WarmStrings`	Plurals are recognised.
`LFO MADNESS`	`LFO Madness`	Known abbreviations (FM, LFO, PWM, EP…) stay in capitals.
`SUBBASS-DIST`	`SubBass-Dist`	Punctuation and digits are kept.
`Kepler`	`Kepler`	Words with lowercase letters are left alone.
`BOFJUPITER`	`Bofjupiter`	Unknown words can't be split; edit the suggestion.

Splitting uses a built-in list of about 100 sound words (pad, lead, bass, string, dist, warm…). Anything else is just given a capital letter, so check the suggestions before confirming.

6 Layer B

Each Prophet 12 program has two layers, A and B. Layer B is heard when **Split A | B** (B plays above the split point) or **Stack A + B** (both play together) is on; otherwise **Edit Layer B** on the synth plays it.

To put any patch into Layer B of a program, drag it onto the **Layer B** column of that program's row in the Prophet 12 pane. Only the Layer B cell is outlined while you hover over it. The dragged patch's Layer A sound and name become the program's Layer B; Layer A stays exactly as it is on the synth.

This is the synth's **Layers → Copy A -> B**, but with the sound taken from any patch. Dropping a patch onto its **own** Layer B does exactly what Copy A -> B does.

Example: a layered pad

1. Read bank `U1` and open a file with a string patch in the File pane.
2. Drag the string patch onto the **Layer B** cell of your pad, e.g. `U1:20`.
3. Confirm. The pad keeps its sound in Layer A and gets the strings in Layer B.
4. On the synth, select `U1:20` and press **Stack A + B** to hear both at once.

> **Note**
> Dropping a patch on the **rest** of a row writes the whole patch to that slot instead. Watch which part of the row is outlined before you let go.

7 Initializing slots

Right-click one or more User slots and choose **Initialize…** to replace them with the synth's Basic Patch, named **Init Patch**: a clean starting point for a new sound.

The first time, the app needs a copy of the Basic Patch, and the dialog explains how:

1. On the synth, press **Global** and go to **23 Basic Patch**.
2. Press **Write Now** (soft key 1). This loads the Basic Patch into the edit buffer, replacing any unsaved edits; nothing is stored.
3. Click **Capture template** in the app.

After that the template is remembered, and **Initialize…** goes straight to the confirmation. **Recapture template…** in the same dialog reads it again.

8 Searching

Type in the search box above either list, or press `Ctrl`+`F` to jump to the box of the pane you used last. `Esc` clears it.

- Upper and lower case don't matter. Several words must **all** match, in any order: `bof pad` finds "BOF COOL PAD".
- The slot counts too: `u2 pad` finds pads in bank U2.
- In the Prophet 12 pane the search covers **every bank that has been read**, not just the open tab. Each tab shows its number of matches; click a tab to clear the search and open that bank. Use **Read all User banks** first to search everything.
- Matching letters are highlighted.

> **Note**
> Patches hidden by a search are never acted on, even if they were selected before. While searching, patches can't be dropped onto the Prophet 12 list, because the slots they would fill might be hidden. Dropping onto a Layer B cell still works.

9 Safety and backups

Every operation that changes the synth (writing, renaming, initializing and Layer B copies) works the same way:

1. The slots about to change are **read from the synth** and saved to a backup file.
2. The new programs are written, one at a time, with a short pause after each so the synth can store it.
3. Every slot is **read back** and compared with what was sent. A mismatch is reported, with the backup's location.

Backups go to `~/Documents/Prophet 12 Backups/` and are named after what was done and when, e.g. `before-rename-20261008-210232.syx`. They are never deleted by the app.

> **Example: undoing a change**
> 1. **Open .syx…** and choose the backup from `~/Documents/Prophet 12 Backups/`.
> 2. Select the patches, click **Write to synth…** and keep **Original locations**.
> 3. Click **Write**.

The Factory banks are never written: the app refuses before anything is sent.

10 Settings

10.1 Zoom, font and theme

`Ctrl`+`+` or `Ctrl` + wheel up	Zoom in
`Ctrl`+`-` or `Ctrl` + wheel down	Zoom out
`Ctrl`+`0`	Actual size

The **View** menu has the same commands.

View → Appearance… (`Ctrl`+`Shift`+`A`) shows every theme as a small preview of the app and every font with a sample line. Click one to use it straight away. The same choices are in **View → Theme** and **View → Font**.

- **Themes:** System (light or dark, following the desktop), Dracula, Catppuccin Mocha, Catppuccin Latte (light), Tokyo Night, Gruvbox Dark, One Dark Pro, Rosé Pine, Kanagawa, Everforest Dark and Solarized Dark.
- **Fonts:** Fira Code, JetBrains Mono and Hack (Nerd Font versions, included with the app; an installed copy is used when present), or the system font.

Zoom (80–250%), font and theme are remembered.

10.2 Where the app keeps things

Backups	`~/Documents/Prophet 12 Backups/`
Settings	`~/.config/app.p12librarian/settings.json`

Init template	`~/.local/share/app.p12librarian/init-template.syx`

11 Keyboard shortcuts

Keys	Action
Ctrl+O	Open a `.syx` file
Ctrl+B	Back up all User banks
Ctrl+F	Search in the pane used last
F2	Rename the selected patches
F1	Open this guide
Ctrl++ Ctrl+- Ctrl+0	Zoom in, out, actual size
Esc	Close a dialog or menu; clear a search
Ctrl+Q	Quit

12 Troubleshooting

"The ports opened, but no Prophet 12 answered" Check **MIDI SysEx Enable** and **MIDI SysEx Cable** in the synth's Global menu (Section 2.1), and that the selected ports are the synth's.

Timeouts or failed verification Some MIDI interfaces corrupt long SysEx messages. Owners have reported problems with Native Instruments Maschine MIDI ports, Digidesign Mbox and CME U2MIDI interfaces. Connect the synth directly by USB.

The app doesn't find the synth after plugging it in Click ↻ next to the port lists.

A patch sounds different after auditioning Auditioning replaced the edit buffer. Select the program on the synth again to reload it.

Layer B can't be heard Turn on **Split A | B** or **Stack A + B** on the synth, or press **Edit Layer B** to play Layer B alone.

13 Command-line tool

The project also includes `p12`, a command-line version of the main functions, which is useful for scripts and scheduled backups. Run `p12 --help` for every option.

```
p12 ports                                    # list MIDI ports
p12 identify                                 # check the synth answers
p12 backup                                   # all User banks to one file
p12 backup -s U1,U2:10-20 -o mine.syx        # a selection
p12 backup -s user --split -o patches/       # one file per program
p12 edit-buffer -o current.syx               # the sound being edited
p12 list mine.syx                            # list a file (no synth needed)
p12 send mine.syx --select U1:05 --audition  # play without storing
p12 send mine.syx --select U1:05 --to U4:99 --verify
```

`send` backs up the slots it will overwrite, shows what will be replaced and asks before writing.

14 Technical notes

For the curious, and for anyone building on this work.

- **Messages** (manual, Appendix D): request a program with `F0 01 2A 05 <bank> <program> F7`; the reply and writes use `F0 01 2A 02 <bank> <program> <1171 bytes> F7`. The edit buffer uses types `06` (request) and `03` (dump). Banks 0–3 are U1–U4 and 4–7 are F1–F4; programs are 0–98.
- **Packing:** each program is 1024 bytes, sent as 1171 MIDI bytes: every 7 bytes travel as 8, with the top bits collected in a leading byte.
- **Layout** (found on the tested synth; not in the manual): Layer A is bytes 0–511 and Layer B bytes 512–1023. The name is at bytes 402–421 (Layer A) and 914–933 (Layer B), 20 characters padded with spaces. Byte positions are **not** the NRPN parameter numbers.
- **Copy A -> B** on the synth copies all 512 bytes of the layer. Bytes 23 and 293 of each layer hold the same value in both layers of every program examined, so they are kept when a layer is copied from another program.
- The **Device Inquiry** reply only reports the main OS version (1 for OS 1.4.2.3); the Global screen shows the full version.

Source code: https://github.com/QutritLogic/P12-Librarian

15 About



P12 Librarian was created by **Claude** (Opus 5.5), an AI model made by Anthropic, who wrote the code, together with **Qutrit**, who had the idea, made the design decisions and tested everything on a real Prophet 12. Contact: qutritlogic@proton.me.

The **Prophet 12** was designed by **Dave Smith Instruments (DSI)**, founded by Dave Smith, and released in 2013. The company is now called Sequential: https://sequential.com/product/prophet-12-keyboard/.

This project is not affiliated with Sequential or Dave Smith Instruments. Prophet 12 is a trademark of its owner. Released under the MIT license.

16 Disclaimer

1. Use at your own risk

P12 Librarian (the "Software") reads and writes data on a hardware synthesizer over MIDI System Exclusive (SysEx). Writing, initializing, renaming or copying programs replaces the data stored in the affected memory locations of the synthesizer. Defects in the Software, the computer, the operating system, USB or MIDI drivers, cables or interfaces, the synthesizer's firmware, interruptions of power or communication, or incorrect use can cause programs, banks, settings or other data on the synthesizer or the computer to be overwritten, corrupted or permanently lost, and may leave the instrument in a state that requires a factory reset, a firmware reinstallation or service. You use the Software entirely at your own risk.

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